Diabetic Retinopathy Analyzer

Project Overview

Overview

A Flask web application that analyzes retinal fundus images and classifies diabetic retinopathy (DR) into five severity stages. It presents the predicted stage, confidence, class probabilities, urgency, and suggested next steps.

Problem Addressed

Diabetic retinopathy can cause vision loss when it is not detected and followed up in time. This project demonstrates how image classification can support screening by organizing a model result with a stage-based action plan.

How It Works

1. The user uploads a retinal fundus image and may enter patient details. 2. The application preprocesses the image. 3. A trained deep-learning model predicts one of five classes. 4. The web page displays the result, confidence, probabilities, urgency, and recommendations. 5. A downloadable PDF clinical report summarizes the prediction and action plan.

Main Features

Image upload and preprocessing; five-stage DR classification; confidence score and complete probability breakdown; stage-specific clinical recommendations and urgency; downloadable PDF clinical report; browser-based interface.

Technology

Python and Flask (web application/API); TensorFlow/Keras (model inference); OpenCV and preprocessing utilities (image preparation); HTML, CSS, and JavaScript (web interface); ReportLab (PDF report generation).

Demonstration Flow

Upload a sample retinal image, review the predicted stage and confidence, discuss the urgency and recommendations, then download the generated PDF report.

Important Limitation

This is an educational and decision-support project. Model predictions and generated recommendations do not replace diagnosis or treatment decisions by a qualified eye-care professional.

Classification Stages

Class	DR Stage
0	No diabetic retinopathy
1	Mild non-proliferative DR
2	Moderate non-proliferative DR
3	Severe non-proliferative DR
4	Proliferative DR